

MADHUR GAHLOT

Computer Science & Engineering | AI/ML & Backend Developer

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EDUCATION

B.Tech | Computer Science & Engineering — Gurukula Kangri Vishwavidyalaya, Haridwar | 2023–2027

EXPERIENCE

C++ Programming Intern | CodSoft

- Developed multiple C++ applications as part of the programming internship.
- Built projects including Calculator, Number Guessing Game, To-Do List, and Tic-Tac-Toe.
- Strengthened programming and problem-solving skills through practical implementation.

PROJECTS

REFLECTAI — User-Authenticated Journal & Gemini Assistant | TypeScript, Firebase, Google Cloud, Gemini API

- Built a secure journaling and AI reflection workspace with authenticated users and multi-turn AI interactions.
- Implemented Google Sign-In, journal threads, autosave, history/search, tagging, Markdown export and AI-assisted reflection workflows.
- Integrated server-side Gemini access with fallback handling and protected secret management for deployment.

Assignment Similarity & AI-Based Grading System | Python, FastAPI, PostgreSQL

- Developed a web-based system for managing assignments and analyzing similarity between student submissions.
- Implemented JWT authentication, assignment management, submission handling and REST APIs.
- Integrated OCR and NLP techniques for typed and handwritten assignments using TrOCR and semantic similarity with Sentence Transformers.
- Implemented automated grading functionality and documented API workflows with Swagger.

COD Tournament Management System | Python, Flask, MySQL — Deployed

- Developed and deployed a web-based tournament management platform.
- Implemented modules for teams, players, fixtures, points tables, tournament updates and player comparison.
- Added login and registration functionality; deployed a live demonstration.

EEYE — Software-Based Air Surveillance Radar | C++, Raylib — In Progress

- Developing a software-based radar simulation for real-time air-surveillance visualization.
- Implemented radar display, grid, sweep animation, range rings and modular C++ architecture.
- Currently expanding target visualization and real-time radar features.

TECHNICAL SKILLS

- **Languages:** C++, Python, Java, JavaScript, SQL, HTML, CSS
- **AI/ML:** Scikit-learn, Pandas, NumPy, NLP, Sentence Transformers, OCR
- **Backend:** FastAPI, Flask, REST APIs, JWT Authentication, SQLAlchemy
- **Databases:** PostgreSQL, MySQL, Firebase/Firestore
- **Tools:** Git, GitHub, VS Code, Docker
- **Graphics:** Raylib
- **Operating Systems:** Windows, Linux

CERTIFICATIONS

AI Literacy — IBM • Getting Started with Artificial Intelligence — IBM • Cloud Computing Fundamentals — IBM • SQL (Basic) — HackerRank • Python (Basic) — HackerRank • C++ Programming — CodSoft • Computer Architecture — NPTEL • Pandas — Kaggle

PROFILES

GitHub: github.com/MadhurGahlot | LinkedIn: linkedin.com/in/madhur-gahlot-a9b614298 | HackerRank: hackerrank.com/profile/madhurgahlot20